

DMZ Configuration Report with Cisco Packet Tracer

1. Lab Objective

Briefly explain what was intended to be achieved with this lab.

Example: Configure a secure Demilitarized Zone (DMZ) using a Cisco ISR router, applying NAT and ACLs to control traffic between the internal LAN network, the DMZ network (containing a public web server), and the external/internet network. The goal is to:

- Isolate critical internal resources from the DMZ
 - Allow controlled external access to web services only
 - Prevent lateral movement (compromised DMZ cannot attack internal LAN)
 - Implement defense-in-depth security using IP addressing, NAT, and firewall rules
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2. Implemented Topology

Describe the network. You can include an image if the software allows it (Packet Tracer screenshot).

Topology Description:

The lab uses a three-zone network segmentation model:

- **Central Router (Router_FW):** Cisco ISR 2911 acting as the security boundary between zones
 - GigabitEthernet0/0: Connects to Internal LAN (trusted)
 - GigabitEthernet0/1: Connects to DMZ (untrusted but controlled)
 - GigabitEthernet0/2: Connects to External/Internet (untrusted)
- **Three Network Segments:**
 1. **Internal LAN (SW_Internal):** Internal users, represents corporate workstations
 2. **DMZ (SW_DMZ):** Public-facing web server, isolated from internal network

3. External (SW_External): Simulates Internet clients accessing services

- **End Devices:**

- PC_Internal: Represents internal user (LAN)
- Server-PT Web_DMZ: Public web server (DMZ)
- PC_External: Represents Internet client (External)

This three-zone design follows the industry standard DMZ architecture, where public services are separated from internal infrastructure by network segmentation and firewall rules.

3. IP Addressing Plan

Complete the table with the assigned IPs (you can copy it from the statement if it hasn't changed).

Device	IP Address	Subnet Mask	Gateway	Zone
PC_Internal	192.168.1.10	255.255.255.0	192.168.1.1	Internal LAN
Server-PT Web_DMZ	192.168.2.10	255.255.255.0	192.168.2.1	DMZ
PC_External	192.168.3.10	255.255.255.0	192.168.3.1	External
Router_FW Gi0/0 (LAN)	192.168.1.1	255.255.255.0	N/A	Internal LAN
Router_FW Gi0/1 (DMZ)	192.168.2.1	255.255.255.0	N/A	DMZ
Router_FW Gi0/2 (Ext)	192.168.3.1	255.255.255.0	N/A	External

NAT Mapping:

- Internal DMZ server (192.168.2.10) is mapped to external router IP (192.168.3.1)
- External clients see the web server at 192.168.3.1, but it's actually at 192.168.2.10 internally

4. Applied Configuration (Summary)

Summarize the most relevant commands or steps you executed. Use text + code snippets

as needed.

Step 1: Router Interface Configuration

Configured three interfaces on Router_FW with appropriate IP addresses:

```
Router_FW(config)# interface GigabitEthernet0/0
Router_FW(config-if)# ip address 192.168.1.1 255.255.255.0
Router_FW(config-if)# no shutdown
Router_FW(config-if)# exit
```

```
Router_FW(config)# interface GigabitEthernet0/1
Router_FW(config-if)# ip address 192.168.2.1 255.255.255.0
Router_FW(config-if)# no shutdown
Router_FW(config-if)# exit
```

```
Router_FW(config)# interface GigabitEthernet0/2
Router_FW(config-if)# ip address 192.168.3.1 255.255.255.0
Router_FW(config-if)# no shutdown
Router_FW(config-if)# exit
```

Step 2: NAT Configuration

Designated inside/outside interfaces and created static NAT mapping:

```
Router_FW(config)# interface GigabitEthernet0/0
Router_FW(config-if)# ip nat inside
Router_FW(config-if)# exit
```

```
Router_FW(config)# interface GigabitEthernet0/1
Router_FW(config-if)# ip nat inside
Router_FW(config-if)# exit
```

```
Router_FW(config)# interface GigabitEthernet0/2
Router_FW(config-if)# ip nat outside
Router_FW(config-if)# exit
```

```
Router_FW(config)# ip nat inside source static 192.168.2.10 192.168.3.1
Router_FW(config)# end
```

Effect: DMZ server (192.168.2.10) is accessible from external network as 192.168.3.1.

Step 3: ACL Configuration (Firewall Rules)

Created two access control lists to implement security policies:

ACL 1 - WAN_TO_DMZ (Internet → DMZ):

```
Router_FW(config)# ip access-list extended WAN_TO_DMZ
Router_FW(config-ext-nacl)# permit tcp any host 192.168.3.1 eq 80
Router_FW(config-ext-nacl)# deny ip any 192.168.2.0 0.0.0.255
Router_FW(config-ext-nacl)# permit ip any any
Router_FW(config-ext-nacl)# exit
```

ACL 2 - DMZ_TO_LAN (DMZ → LAN):

```
Router_FW(config)# ip access-list extended DMZ_TO_LAN
Router_FW(config-ext-nacl)# deny ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router_FW(config-ext-nacl)# permit ip any any
Router_FW(config-ext-nacl)# exit
```

Step 4: Apply ACLs to Interfaces

Attached ACLs to router interfaces in the inbound direction:

```
Router_FW(config)# interface GigabitEthernet0/2
Router_FW(config-if)# ip access-group WAN_TO_DMZ in
Router_FW(config-if)# exit

Router_FW(config)# interface GigabitEthernet0/1
Router_FW(config-if)# ip access-group DMZ_TO_LAN in
Router_FW(config-if)# exit

Router_FW(config)# end
Router_FW# write memory
```

Step 5: Web Server Configuration

Activated HTTP service on Server-PT Web_DMZ:

- Services tab → HTTP: **ON**

5. Verifications Performed

Describe the tests and their results. Include screenshots or command outputs if possible.

Test 1: Baseline Connectivity - Ping to Gateways

Objective: Verify basic connectivity from each zone to its gateway before applying security

rules.

Commands Executed:

Source	Command	Expected Result	Actual Result
PC_Internal	ping 192.168.1.1	4 replies, 0% loss	✅ PASS - 4/4 replies
Server Web_DMZ	ping 192.168.2.1	4 replies, 0% loss	✅ PASS - 4/4 replies
PC_External	ping 192.168.3.1	4 replies, 0% loss	✅ PASS - 4/4 replies

Analysis: All devices can reach their respective gateways. Network baseline is functional.

Test 2: NAT & Web Access (Pre-ACL)

Objective: Verify that NAT translation works and web services are accessible before ACLs are applied.

Commands Executed:

Source	Action	Target	Expected Result	Actual Result
PC_External	Web Browser	http://192.168.3.1	Page loads	✅ PASS - Cisco Packet Tracer home page loaded
PC_Internal	Web Browser	http://192.168.2.10	Page loads	✅ PASS - Cisco Packet Tracer home page loaded

Analysis: NAT is working correctly. External clients see the server at 192.168.3.1, internal clients can reach it at 192.168.2.10. Both paths work as expected.

Test 3: Security - DMZ to LAN Blocking

Objective: Verify that ACL blocks lateral movement from DMZ to internal LAN.

Command Executed (Server Web_DMZ):

```
C:\> ping 192.168.1.10
```

```
Pinging 192.168.1.10 with 32 bytes of data:
```

```
Reply from 192.168.2.1: Destination host unreachable.  
Reply from 192.168.2.1: Destination host unreachable.  
Reply from 192.168.2.1: Destination host unreachable.  
Reply from 192.168.2.1: Destination host unreachable.
```

```
Ping statistics for 192.168.1.10:
```

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Result:  **PASS** - ACL successfully blocks DMZ → LAN traffic

Analysis: The DMZ_TO_LAN ACL is functioning correctly. If a server in the DMZ is compromised, the attacker cannot pivot to internal users.

Test 4: Security - LAN to DMZ Blocking

Objective: Verify that internal LAN also cannot communicate with DMZ (bilateral isolation).

Command Executed (PC_Internal):

```
C:\> ping 192.168.2.10
```

```
Pinging 192.168.2.10 with 32 bytes of data:
```

```
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.
```

```
Ping statistics for 192.168.2.10:
```

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Result:  **PASS** - ACL successfully blocks LAN → DMZ traffic

Analysis: Bilateral isolation between DMZ and LAN is complete. This prevents internal users from accidentally or intentionally accessing DMZ administration interfaces.

Test 5: Security - HTTP from External (Allowed)

Objective: Verify that external clients can still access the web server despite ACLs.

Action (PC_External):

Web Browser → http://192.168.3.1

Result: Page loads successfully

Result:  **PASS** - HTTP from external network is permitted

Analysis: The WAN_TO_DMZ ACL correctly allows TCP port 80 (HTTP) to the DMZ server while blocking other protocols.

6. Conclusions and Recommendations

What did you learn from this exercise? What would you improve?

Key Learnings

1. **Network Segmentation Works:** By dividing the network into three zones (Internal, DMZ, External) and controlling traffic between them, we can significantly reduce security risk. If the DMZ server is compromised, the attacker is contained.
2. **NAT is Powerful:** Static NAT allows us to expose a private IP address to the Internet without revealing internal infrastructure details. The external network sees 192.168.3.1, but the actual server is at 192.168.2.10 and completely hidden.
3. **ACLs are Simple but Effective:** Even simple “allow HTTP, deny everything else” rules provide strong protection. The order of rules matters (first match wins), and implicit deny is the default behavior.
4. **Defense in Depth:** Using IP addressing + NAT + ACLs together creates layered security:
 - IP addressing segregates traffic
 - NAT hides infrastructure
 - ACLs enforce policies
5. **Bilateral Isolation is Important:** We blocked not just DMZ→LAN, but also LAN→DMZ. This prevents internal users from accidentally exposing themselves to DMZ services.

Recommendations for Improvement

- **Stateful Firewall:** Real firewalls (like Cisco ASA) use stateful inspection—they remember outbound connections and allow replies automatically. Standard ACLs are stateless.
- **Logging:** Enable logging on ACLs to see what traffic is being blocked: `access-list 101 permit ... log`

- **SSH Access:** Use SSH instead of Telnet for router management. Telnet is unencrypted.
- **VLAN Segmentation:** Extend this model using VLANs on switches for additional network isolation.
- **IDS/IPS:** Add intrusion detection/prevention systems in the DMZ for threat detection.

Real-World Applicability

This three-zone DMZ design is used in:

- **E-commerce:** Customer-facing web servers isolated from order databases
- **Banking:** ATM interfaces in DMZ, core banking systems on protected LAN
- **Government:** Classified networks use even stricter segmentation
- **Hospitals:** Patient data protected behind DMZ-like architecture

The principles learned here scale to enterprise networks with dozens of zones and thousands of devices.

7. Evidence Screenshots

Attach here (or in an annexed PDF) the requested screenshots: pings, browser, "show" commands, etc.

Required Screenshots:

1.  **Baseline Connectivity - PC_Internal ping to gateway (192.168.1.1)**
 - Shows: 4 replies, 0% loss
2.  **Baseline Connectivity - Server ping to gateway (192.168.2.1)**
 - Shows: 4 replies, 0% loss
3.  **Baseline Connectivity - PC_External ping to gateway (192.168.3.1)**
 - Shows: 4 replies, 0% loss
4.  **NAT Test - PC_External Web Browser accessing <http://192.168.3.1>**
 - Shows: Page loads (Cisco Packet Tracer homepage)
5.  **Security Test - Server Web_DMZ ping 192.168.1.10**

- Shows: "Destination host unreachable" (100% loss)

6. **Security Test - PC_Internal ping 192.168.2.10**

- Shows: "Request timed out" (100% loss)

7. **Router Configuration Verification**

- Shows: `show running-config` or ACL summary

Lab Completion Status:  COMPLETE

All configuration steps executed successfully. All security tests passing. Report submitted with evidence.

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